

Name:

ADM No:

Section:

SEMESTER: MONSOON, SESSION: 2022-2023

Quiz-1

Subject: Basics of Electronics Engineering

Subject Code: ECI101

Time: 30 Mins

Max Marks: 26

Instructions: **Tick only the correct option.**

1. The peak inverse voltage rating of a diode in a bridge rectifier is _____ times that of a center tap full wave rectifier yielding the same dc output voltage. [1 Mark]
(A) 2 (B) 1 (C) $\sqrt{2}$ (D) 0.5
2. The circuit shown in Fig.1 contains ideal diodes and capacitors. Find the voltage across the diode D_1 if the input sinusoid voltage V_1 is $\cos(\omega t)$. [2 Mark]
(A) $\cos(\omega t)$ (B) $1 - \cos(\omega t)$ (C) $-1 + \cos(\omega t)$ (D) None of these
3. Find the dynamic, ie AC resistance of a diode having $V_D = 0.2$ V and $I_D = 2.27$ mA. [1 Mark]
(A) 0.0Ω (B) 11.4Ω (C) 30.8Ω (D) 88.1Ω
4. Find the values of I and V for the circuit given in Fig.2 [2 Mark]
(A) $I = 1$ mA, $V = 1$ V (B) $I = 2$ mA, $V = 2$ V (C) $I = 3$ mA, $V = 3$ V (D) Indeterminate
5. For a half-wave rectifier, the value of ripple factor is [1 Mark]
(A) 2 (B) 1.21 (C) 1.57 (D) 1
6. Assume that the diodes in the circuit in Fig.3 to be ideal. For the output to be clipped, the input voltage V_i must be outside the range [2 Mark]
(A) -1 V to -2 V (B) -2 V to -4 V (C) -2 V to +1 V (D) -4 V to +2 V
7. Find the approximate change in diode voltage if the current changes from 0.1 mA to 10 mA. Assume the diode ideality factor $\eta = 1$. [2 Mark]
(A) 100 mV (B) 120 mV (C) 9.9 mV (D) 100 V
8. A full wave rectifier is fed with a 50Hz sine wave. What is the ripple frequency at the output? [1 Mark]
(A) 25Hz (B) 50Hz (C) 100Hz (D) 0 Hz
9. A crystal diode having internal resistance $r_f = 20\Omega$ is used for half-wave rectification. If the applied voltage $V_i = 50 \sin \omega t$ and load resistance $R_L = 800 \Omega$, find : [2 Mark]
(i) D.C output voltage (ii) efficiency of rectification.
(A) 50V, 40.6% (B) 15.52 Volt, 39.5% (C) 50 Volt, 39.5% (D) 15.52 Volt, 40.6%
10. Determine I_D and V_D For the circuit given in Fig.4. Assume the diode is a Si diode with cut-in voltage of 0.7V. [2 Mark]
(A) $I_D = 0$, $V_D = 0.7$ V (B) $I_D = 0.015$ mA, $V_D = 0.7$ V (C) $I_D = 0.01$, $V_D = 0.7$ V (D) $I_D = 0$, $V_D = 0.45$ V
11. A half-wave rectifier is used to supply 50V d.c. to a resistive load of 800Ω . The diode has a resistance of 25Ω . Calculate the maximum a.c. voltage required. [1 Mark]
(A) 162V (B) 81V (C) 50V (D) 51.5625V
12. The form factor of a rectifier is defined as the ratio of [1 Mark]

- (A) RMS value : Average Value
 (B) RMS value : Peak Value
 (C) Peak Value : Average Value
 (D) Average Value : RMS value

13. Consider the circuit shown in Fig.5. If the input voltage is a sinusoid with 40 V peak to peak, then the output is: [2 Mark]

- (A) Maximum voltage of 7.5 V and minimum voltage of -15 V
 (B) Maximum voltage of 12.5 V and minimum voltage of -15 V
 (C) Maximum voltage of 15 V and minimum voltage of -25 V
 (D) None of these

14. The current flowing through the Zener diode in the Fig. 6 is [2 Mark]

- (A) 20mA (B) 25mA (C) 30mA (D) 5mA

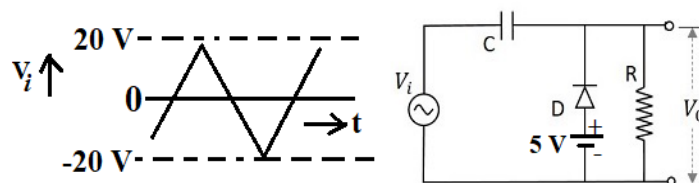
15. A 4.7 V Zener diode has a resistance of 15 Ω . What is the terminal voltage, when the current is 20 mA? [1Mark]

- (A) 0 V (B) 7.5 V (C) 5 V (D) 10 V

16. By narrowing the space charge region, the amount of reverse voltage needed to cause Zener breakdown- [1 Mark]

- (A) Decreases (B) Increases (C) Zero (D) No effect

17. Assume the diode D to be ideal. The maximum and minimum values of the output voltage V_o of the circuit shown in figure below are respectively [2 Mark]



- (A) + 25 V and - 15 V (B) + 15 V and - 25 V (C) + 20 V and - 15 V (D) + 45 V and + 5 V

